

POWER ELECTRONICS

Class B

Push-Pull Amplifier

Problems with Class A Amplifier

- Load is directly connected to source terminal
- Bias current flows continuously even when no signal is applied

$$P_{supply} = 56 \times 3.5 = 196 \text{ W}$$

$$P_{load} = (3.5)^2 \times 8 = 98 \text{ W}$$

- Power dissipation in the transistor

$$P_Q = 196 - 98 = 98 \text{ W}$$

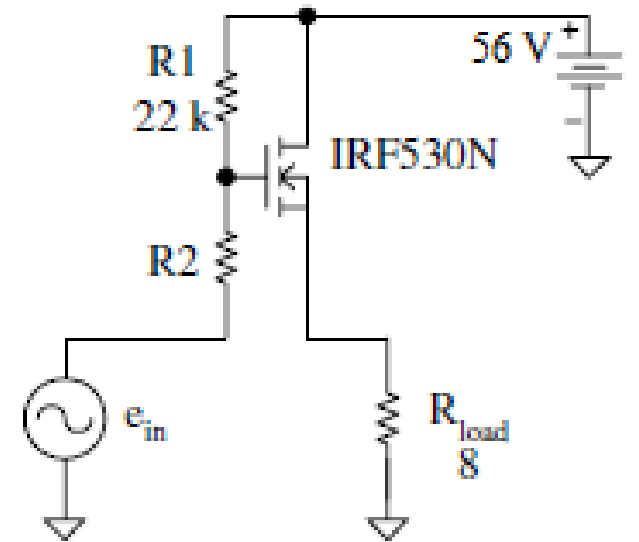
- This power is wasted. It is not produced by the signal. The output (loudspeaker) will burn out even before making any sound
- Class A amplifier is extremely wasteful.
- Rarely used

Class B Amplifier

- Transistor is biased off, just at the verge of conduction i.e. $V_{GS} = V_{th}$
- When input is 0, transistor is off and that 98 W wasted by class A amplifier is not used at all

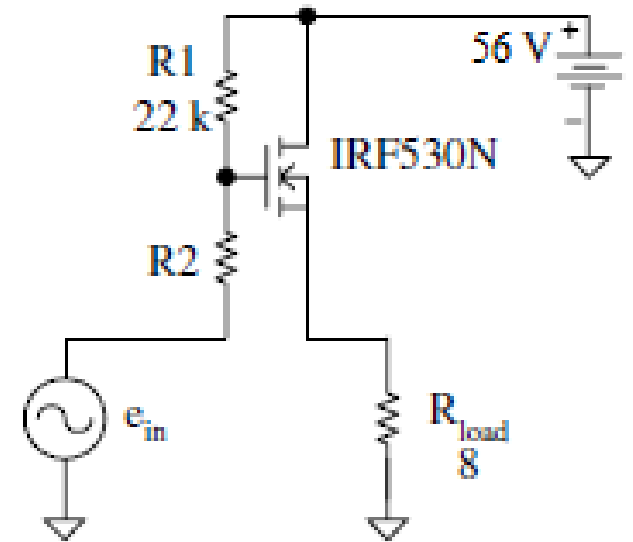
Push Amplifier

- Differences from class A amplifier?



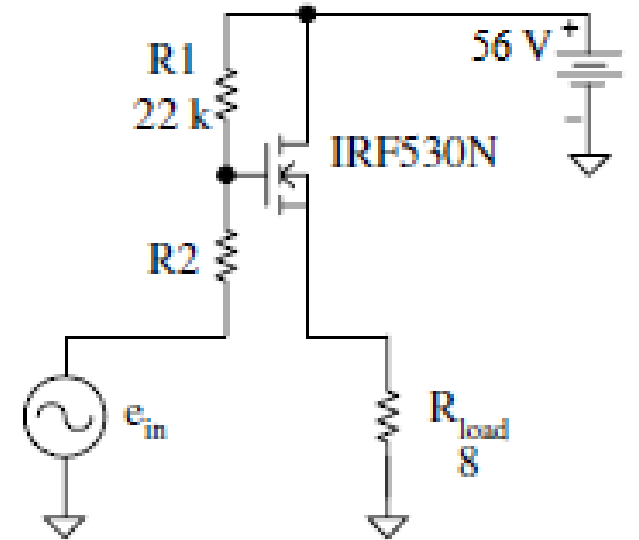
Push Amplifier

- No input capacitor
- Input signal generator drives the lower end of R_2
- Value of R_2 needs to be changed to shift the bias from the middle of the linear region down to the threshold voltage
- Set R_2 to keep $V_{GS} = V_{th}$



Example

- For the circuit shown in figure
 1. Calculate R_2 to establish class B operation. (consider $V_{th} = 3.6$ V)
 2. Determine V_{out} shape and amplitude with $e_{in} = 40$ V peak
 3. Determine P_{load} , P_{supply} and P_Q for the part 2 above.

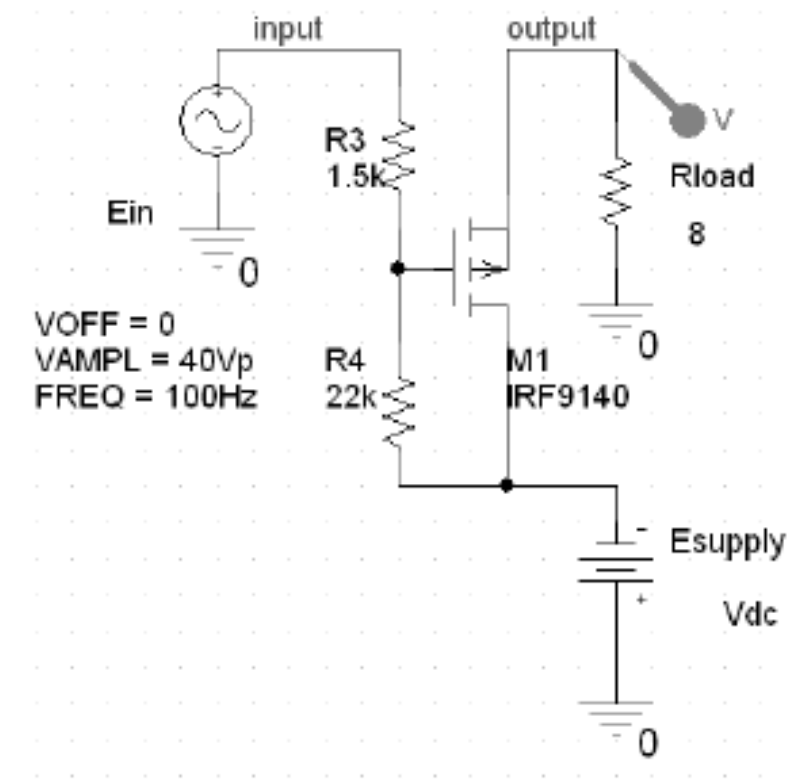


Pull Amplifier

- With push side amplifier, positive signal is sent to load with only a small voltage drop
- High current
- No power is wasted on biasing the transistor
- But the entire negative half of the signal is missing
- On negative side, the n-channel transistor turns off
- A separate pull side is needed to process the negative half of the input
- p -channel MOSFET serves the purpose of pull side amplifier

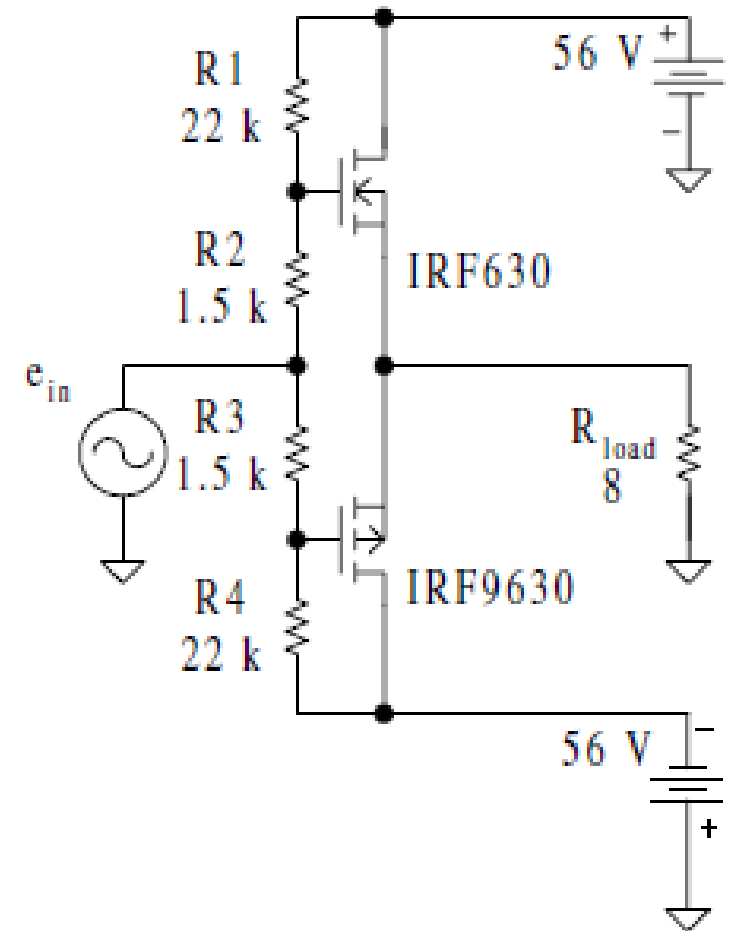
Pull Amplifier

- Same working as push amplifier but processes negative half of the input signal
- As input goes positive, p-channel transistor turns off



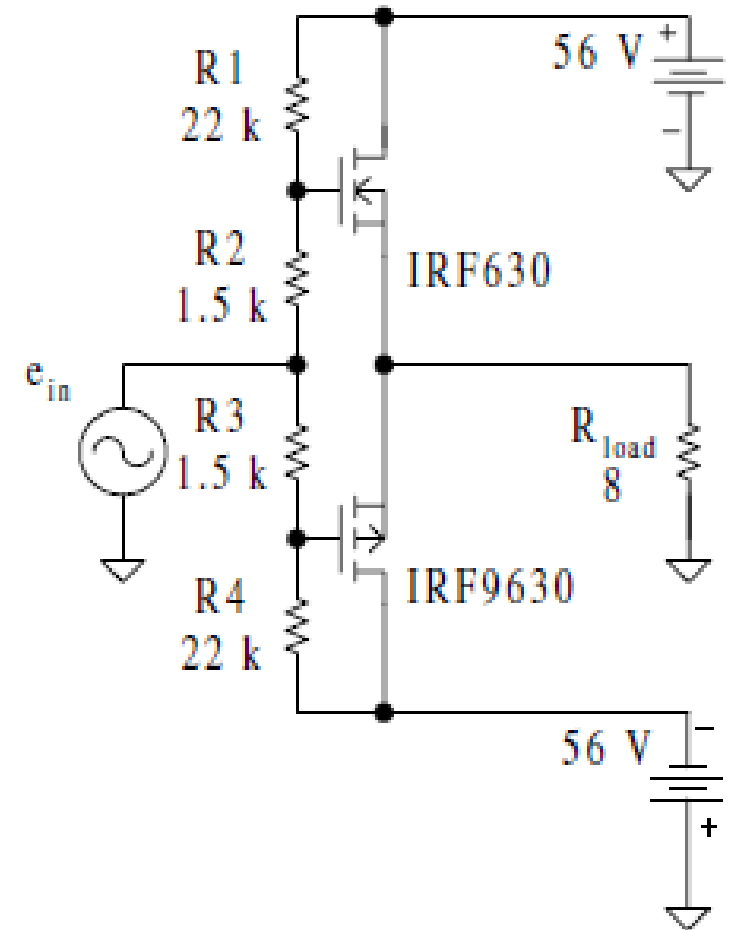
Push-Pull Amplifier

- Each transistor is biased at the edge of conduction but neither is on
- No dc current flows through the load
- No power is dissipated in biasing the transistor
- All calculations remain same as that of previous example



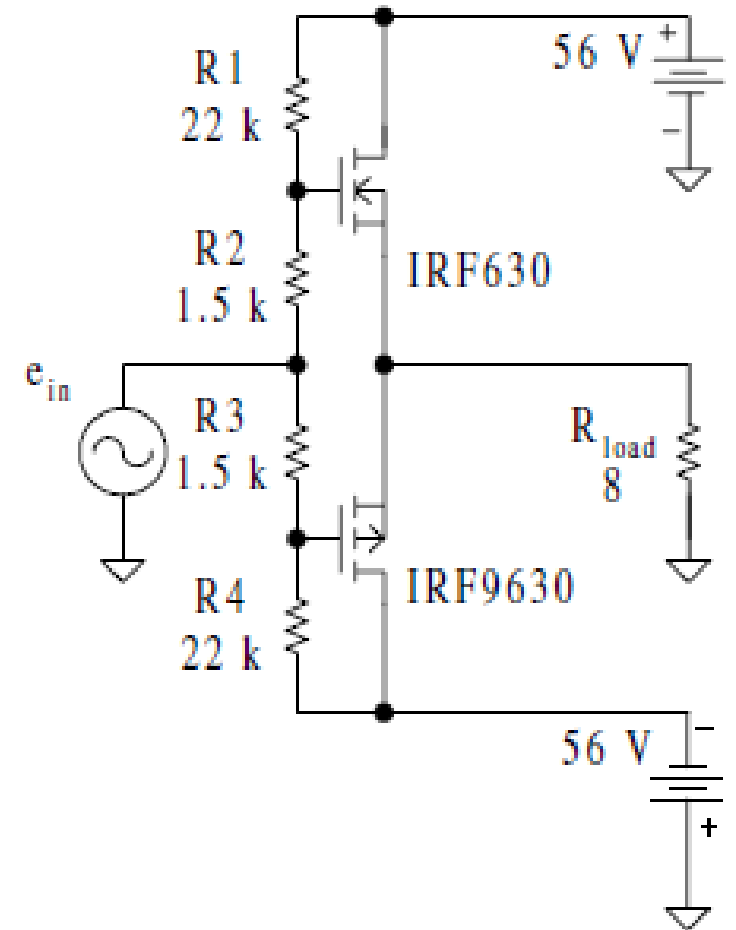
Push-Pull Amplifier

- When input goes positive, n-channel transistor turns on
 - P-channel remains off,
 - 42 W delivers to load (only positive half)
 - Power supply provides 82 W
 - Transistor dissipation is 40 W
- When input goes negative, p-channel transistor turns on
 - N-channel turns off,
 - 42 W delivers to load (only negative half)
 - Power supply provides 82 W
 - Transistor dissipation is 40 W



Push-Pull Amplifier

- Two half signals combine at load, producing a full sine wave
 - 36.5 V peak, 73 V peak-to-peak, 25.8 V rms
 - 84 W
- Each supply delivers 82 W
 - Total power from the supplies is 164 W
- Each transistor dissipates 40 W



Pros of Class B Amplifier

- Class A amplifier is so inefficient
- Impractical for use in delivering more than few watts to a load
- Class B amplifier solves this problem
- Enables delivering hundreds of watts to the load

Cons of Class B Amplifier

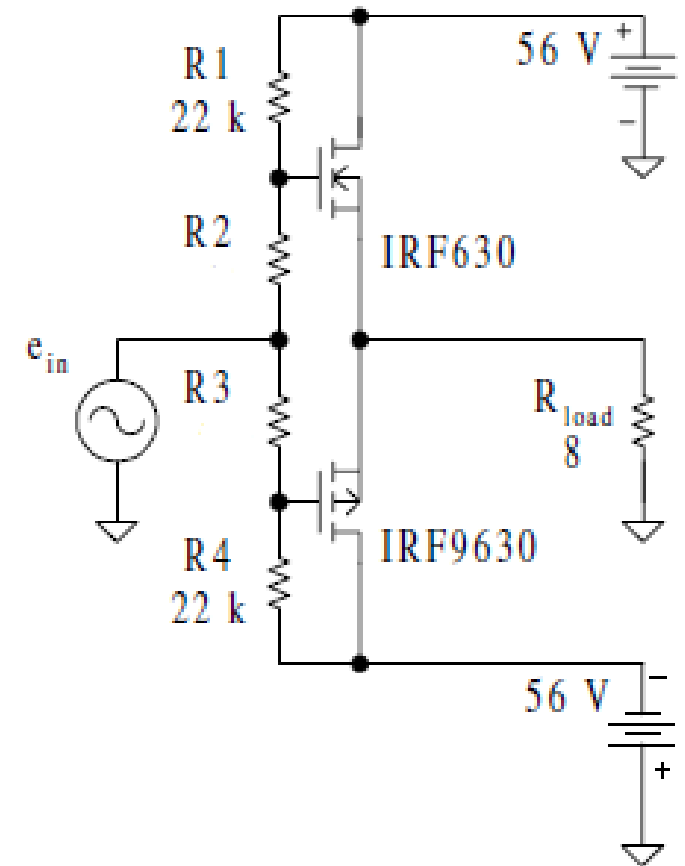
- Since it is a common drain amplifier, gain is always less than 1
- Changes in drain current depends on square of gate-to-source voltage
- Load voltage is created by this drain current
- Therefore, it is a nonlinear relationship between input and output voltage
- The output is distorted
- Larger the signal, the worse the distortion

Cons of Class B Amplifier

- Another issue is the transistor characteristics
- Manufacturer says the transistor turns on at 3.6 V
- But in practical with real transistors, it may turn on at 2.1 V or remain off till 3.9 V
- It is required to measure threshold voltage of every transistor before its deployment in the circuit
- Transistor characteristics depends on temperature, time, humidity, and other unknown factors
- Even a small error or shift in behavior could have a huge effect
- Only safe solution is to change the bias as $V_{GS} = 2 \text{ V}$ (each transistor is off)

Example

- Calculate the new values of R_2 and R_3 to set the gate voltages at ± 2 V



Reference

Power Electronics, Principles and Applications
J. Michael Jacob

Power Electronics
Ned Mohan